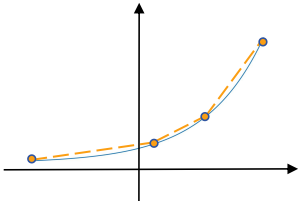


# NLSpike Operator

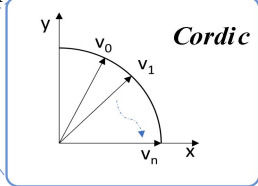
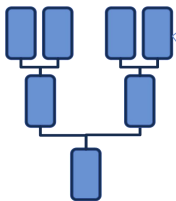
## PWL-EXP Unit



Piecewise Linear Exponential in  $e^x[-5,5]$

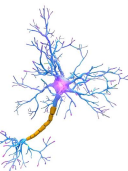
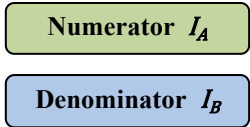


## Division Neural Group



binary-tree

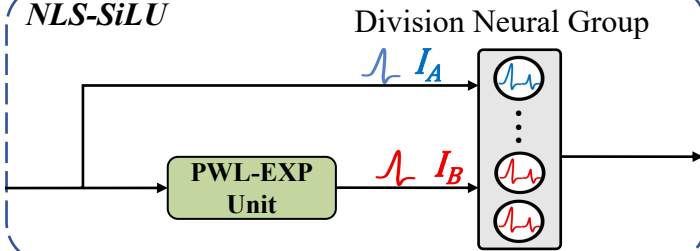
## PolarNorm Unit



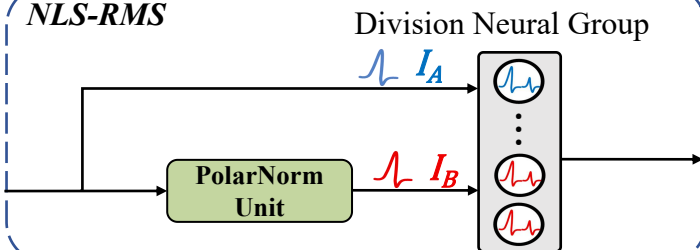
Threshold:  $\theta = I_B \gg n$  Output:  $\hat{q} = q \gg n$

# NLSpike Function

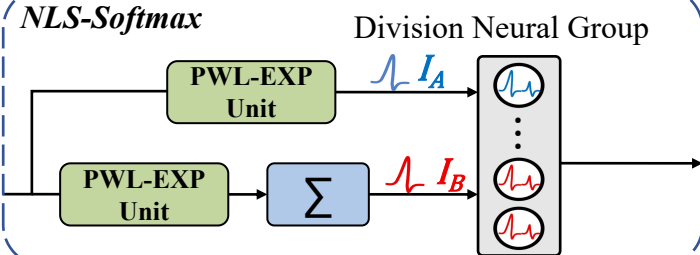
## NLS-SiLU



## NLS-RMS

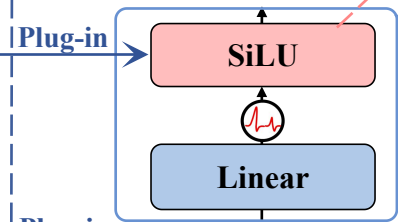


## NLS-Softmax



$$\phi_{SiLU} = \frac{x}{1 + e^{-x}}$$

Division & Exponentiation



$$\phi_{RMS} = \frac{x}{\sqrt{\sum_i x_i^2 + \epsilon}}$$

Mean Square & Division

$$\phi_{Softmax} = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

Division & Exponentiation

